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Predicting first test day milk yield of dairy heifers

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ABSTRACT

Profound physiological changes occur in the body of a dairy cow during the transition period. Most of the metabolic disorders and infectious diseases are diagnosed during this time compromising the lactation overall success. Even though it has been proposed methods to evaluate the transition programme in multiparous dairy cows, the same has not been observed for dairy heifers. The present paper aimed at determining which Dairy Herd Improvement (DHI) metrics have the largest impact on first test day milk yield (FT_MILK) of primiparous dairy cows as well as building and comparing predictive models of FT_MILK from animals in this category. We used information from the first test day of 3267 Holstein primiparous dairy cows collected between 2014 and 2017. Data were split into training (n = 2345) and validation (n = 780) data set that were used to estimate model parameters and to evaluate the models, respectively. Variables associated to FT_MILK were identified using regularized regression via elastic net. Three types of models were evaluated: multivariate linear regression (MLR), random forest (RF), and artificial neural network (ANN). The resulting models were evaluated based on six fit-statistics and 10-fold cross validation. In addition, Pearson correlation coefficient and Lin's concordance correlation coefficient were calculated between predicted and observed values, as well as the comparison between observed and predicted median FT_MILK. Statistical significance was declared at an error level $\alpha < 0.05$. Fifteen DHI variables were identified as the most influential on FT_MILK. The three models were capable of predicting FT_MILK with a mean absolute error lower than 4.00 kg on validation data set and 10-fold cross validation. There was no significant difference between the observed and predicted median FT_MILK. ANN better-predicted FT_MILK compared with MLR and RF. Future work could focus on studying the variation between herds as a possibility to improve the ANN model.

1. Introduction

The transition period of dairy cows, which is the six-week period around parturition, is characterized by profound physiological and endocrinal changes in the body of a dairy cow (Drackley, 1999). Primiparous and multiparous dairy cows reduce their dry matter intake during this time (Grummer et al., 2004) while, simultaneously, there is an increase in nutritional requirements due to the beginning of milk production (NRC, 2001). Therefore, a negative energy balance (NEB) and a dysfunction of the immune system are often observed during the transition period (Drackley, 1999; Lacasse et al., 2018). During this period, animals are more susceptible to develop metabolic disorders, which will negatively affect their milk production and future reproductive performance (Gantner et al., 2016; McArt et al., 2012). Therefore, adequate management of transition cows should aim at reducing the NEB (Lacasse et al., 2018) and avoiding its negative long term effect on productive performance of the animals (Drackley, 1999).

The transition period is as challenging for dairy heifers as to adult cows. While milk production is homeorhetically favored (Drackley and Cardoso, 2014), it competes for nutrients that were once solely directed to body growth in heifers (NRC, 2001). Turk et al. (2013) evaluated the transition period of dairy heifers and reported the existence of a complex adaptation system to the NEB based on the mobilization of fat tissues that stabilizes as the lactation advances. In addition, the occurrence of rumen acidosis (Penner et al., 2007) and ketosis (Santschi et al., 2016) are high soon after calving in primiparous. Therefore, it indicates that primiparous and multiparous dairy cows face similar metabolic challenges and physiological changes during the transition period (Drackley, 1999).

The Transition Cow Index™ (TCI; Nordlund, 2006) was proposed as a means of evaluating the success or failure of the transition period of

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dairy cows. The index is the difference between observed and predicted milk yield at the first test day of a new lactation and it can be expressed as a 305-day milk yield difference (Nordlund et al., 2011). In order to predict milk yield, the TCI model uses parameters obtained from the current and previous lactation (Nordlund, 2006). Therefore, its usage is limited for dairy cows on their second or higher lactation.

Therefore, we believe it is necessary to create a tool to evaluate the transition period of dairy heifers in order to assist technicians and dairy farmers in the process of decision-making. In addition, given that the transition period is similar between primiparous and multiparous dairy cows, a successful strategy deployed for multiparous could be similarly used for primiparous cows. The first step towards that end would be to create a model that predicts first test day milk yield of dairy heifers.

Different modeling methods have been used on dairy data sets. Conventional multivariate linear regression (MLR) is one of the simplest modelling methods available and has been used to predict 305-day milk yield of dairy cows (Murphy et al., 2014). On the other hand, more complex modelling methods such as random forest (RF) and artificial neural networks (ANN) are more flexible and handle non-linearity more easily, which in turn require less programming. For instance, ANN has been successfully used to predict 305-day milk yield of dairy cows (Murphy et al., 2014) as well as to predict calving (Borchers et al., 2017) and calving difficult (Fenlon et al., 2017). On the other hand, RF has been used to identify polymorphisms in nucleotides associated to diet intake of dairy cows (Yao et al., 2013), but it has not been used yet to predict milk the yield of dairy cows.

Therefore, the present paper aimed at determining which Dairy Herd Improvement (DHI) metrics have the largest impact on first test day milk yield (FT_MILK) of primiparous dairy cows as well as building and comparing predictive models of FT_MILK from animals in this category.

2. Materials and methods

Data used in this study was obtained from Valacta – Dairy Production Centre of Expertise in Québec and Atlantic Regions. All data were collected by producers and Valacta technicians as part of the regular Valacta on-farm recording and conformed to normal farm animal handling; consequently, Ethics Committee on the Use of Animals approval was not required.

Data handling and modelling was done using specific packages in R (R Core Team, 2018). Data were refined and cleaned to account for inconsistencies. Outliers were identified and removed using the methodology proposed by Leys et al. (2013). Missing data were imputed using K-nearest neighbors. Regularized regression via elastic net (Zou and Hastie, 2005) was used to select variables associated to FT_MILK. Then, multivariate linear regression (MLR), random forest (RF), and artificial neural network (ANN) were tested as their capability to predict FT_MILK.

2.1. Creating the working data set

The data provided contained 39 variables regarding the test day (TEST), complete lactation (LACTATION), and body weight (WEIGHT) of primiparous Holstein cows from 100 dairy herds located in the province of Quebec – Canada. Apart from the WEIGHT data set, each file contained a single observation per cow collected in 8923 animals from 2014 to 2017.

The WEIGTH data set contained 11,042 observations obtained in 6478 dairy heifers from 2009 and 2017. The body weight was measured and recorded in kilograms by producers and Valacta technicians. Identification and removal of outliers were done following the methodology proposed by Leys et al. (2013), in which the range of valid observations was defined as 2.5 times the median absolute deviation of body weights observations according to the age of the animals in months. Body weight was measured at different ages and different intervals among dairy farms. Therefore, the model developed by Cue et al. (2012) for Holsteins cows was used to estimate the body weight of each animal at 3, 9, 12, and 15 months of age, the average daily gain between theses ages, and the weight at calving.

The TEST data set initially contained 8923 observations collected between January 2014 and October 2017. Observations in which days in milk were higher than 40 days (n = 3755) were removed in order to keep only results from the first test day. In addition, incomplete observations among the different files (n = 1901) were discarded based on animal ID, which was a variable present in all three files and unique to each animal. Lastly, all information was merged into a single file (n = 3267).

Six additional variables were created from existing variables. Season of birth (spring, summer, fall, and winter) and calving (spring, summer, fall, and winter) as well as month of birth and calving were created based on birth and calving date, respectively. Fat to protein ratio was calculated based on test day yield of these milk constituents (kg day^{-1}). A variable indicating the occurrence of twinning was also created.

Missing observations on milk urea nitrogen (n = 710), calving ease (n = 184), calf sex (n = 97), calf size (n = 237), and calf mortality (n = 7) were imputed by means of K-nearest neighbors technique setting k = 5 and using the *KNN* function from the *VIM* package in R (Kowarik and Templ, 2016).

2.2. Setting predictors and elastic net

Variable associated to FT_MILK were identified by submitting the data set to regularized regression via elastic net (Zou and Hastie, 2005). The parameters α, which controls the number of selected variables, and λ, which controls the elasticity of the net, were estimated using 10-fold cross validation testing 100 possible values varying between 0 and 1 using the *cv.glmnet* function from the *glmnet* package in R (Friedma et al., 2010). Next, the *train* function from the *caret* package (Kuhn, 2018) was used to generate the final model specifying the values of α and λ estimated. In the present study, we found the values of 0.46 and 0.52 for α and λ, respectively, and selected 15 variables (Tables 1 and 2).

Variance inflation factors (VIF) were calculated to evaluate the multicollinearity between the 15 variables selected by the regularized regression via elastic net. A threshold of 5.0 was used to evaluate the estimated VIF (James et al., 2013). A bivariate correlation matrix was calculated between the 15 variables and FT_MILK. In addition, autocorrelation between the FT_MILK observations was evaluated using the estimated autocorrelation function for time lags varying from 1 to 35.

The data set used for further analyses contained FT_MILK, which was the target variable, and the 15 variables selected by regularized regression via elastic net (Tables 1 and 2). It was split into training (n = 2452) and validation (n = 815) data sets following a 75 to 25 ratio, respectively. The training data set was used to train all the models and the validation data set was used to evaluate their performance.

2.3. Modeling methods

Multivariate linear regression (MLR), random forest (RF), and artificial neural network (ANN) were the modeling methods used in the present study.

2.3.1. Multivariate linear regression

The training data set was submitted to multivariate linear regression (MLR) using following model: $Y = \alpha_1 X_1 + \alpha_2 X_2 + \ldots + \alpha_n X_n + \varepsilon$, in which Y is the dependent variable, $X_1, X_2, \ldots, X_n$ are the independent variables, $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the regression coefficients, and ε is the residual error. In this study, FT_MILK was the dependent variable and the 15 variables selected by the regularized regression via elastic net were the independent variables (Tables 1 and 2).

The normality of the residual error was assessed using the Shapiro-Wilk test and the QQ-Normal plot. Homoscedasticity was evaluated using the Breusch-Pagan test and plotting the standardized residuals versus fitted values. Furthermore, the independence of the residual error was evaluated using the Durbin-Watson test. Statistical significance was declared at an error level of 5% of probability.

Table 1
Descriptive statistics of the target variable (milk yield on first test day) and numeric variables selected by regularized regression via elastic net.

Variable	Training[a] (n = 2452)	Validation[a] (n = 815)
Milk yield on first test day (kg)	29.5 ± 6.03 29.6 10.0–53.2	29.4 ± 6.09 29.5 11.5–55.6
Days in milk[b]	21.7 ± 9.77 22.0 5–40	21.7 ± 9.72 21.0 5–40
Milk urea nitrogen (mg dL^{-1})[c]	10.6 ± 3.04 10.5 0.0–36.5	10.7 ± 2.87 10.6 2.6–19.8
Somatic cell count (10^3 cells mL^{-1})[c]	159.9 ± 518.13 49.0 1.0–9999.0	148.9 ± 439.78 46.0 3.0–6379.0
Somatic cell count linear score[c]	2.3 ± 1.60 2.0 0.1–9.6	2.2 ± 1.56 1.9 0.1–9.0
Milk β-hydroxybutyrate (mmol L^{-1})[c]	0.10 ± 0.06 0.09 0.0–0.77	0.10 ± 0.07 0.09 0.0–1.10
Fat to protein ratio[c]	1.33 ± 0.27 1.30 0.33–3.96	1.32 ± 0.27 1.28 0.51–3.33
Body weight at calving (Kg)[d]	615.9 ± 63.57 614.8 454.8–988.0	618.3 ± 61.12 614.2 436.1–898.3
Average daily gain between 12 and 15 months old (Kg $animal^{-1}$ day^{-1})	0.750 ± 0.09 0.750 0.520–1.110	0.760 ± 0.09 0.750 0.530–1.000
Calving month[e]	7.0 ± 3.56 7.3 1.0–12.9	7.2 ± 3.49 7.4 1.0–12.9

[a] Mean ± standard deviation; median; range.
[b] Days in milk at first test day.
[c] Measured on first test day.
[d] Estimated using a mixed effect model developed by Cue et al. (2012), which was developed based on body weights of Holstein dairy heifers from Quebec, Canada.
[e] Take into account month and day in which calving occurred. It ranges from 1.0 to 12.9 indicating calving occurrence on the first day of January and on the last day of December, respectively.

2.3.2. Random forest

Random forest (RF) consists of the combination of decision trees r_X (x, $β_k$), k = 1, 2, …, n, where $β_k$ is the subset of variables randomly selected and used as the recursive partition criterion for individual trees (Breiman, 2001a). In this way, the hyperparameters number of trees and βk must be defined. The optimum value of $β_k$ was determined as $β_k$ = 5 using 10-fold cross validation. The number of trees was set at 2000, since it should be large enough to allow each observation to be predicted more than once during the training process (Breiman, 2001a). The training data set, which was composed by FT_MILK and the 15 variables selected by the regularized regression via elastic net (Tables 1 and 2), was used to determine $β_k$ and to train the RF model.

2.3.3. Artificial neural network

Categorical variables from the training and validation data sets (Table 2) were one-hot encoded before submitted to ANN. Next, numerical (Table 1) and categorical (Table 2) independent variables were normalized to mean = 0 and standard deviation = 1.

In the present study, grid search was used to find the optimum architecture of the ANN. In total, 288 different ANN were trained, resulting from a unique combination of the number of hidden layers (1, 2, 3, or 4), number of neurons in each hidden layer (50, 100, or 200), dropout ratio [0, 25 or 50%], L1 (0 or 1×10^{-3}) and L2 (0 or 1×10^{-3}) regularization factors, and activation functions (exponential linear unit or hyperbolic tangent). L1 and L2 regularization factors as well as the dropout

Table 2
Distribution of categorical variables selected by regularized regression via elastic net in the training and validation data sets.

Variable	Training (n = 2452)		Validation (n = 815)	
	N	%	N	%
Milk sampling[a]				
24 h	1505	61.38	504	61.84
Morning	460	18.76	144	17.67
Afternoon	444	18.11	156	19.14
Other	43	1.75	11	1.35
Milking frequency[b]				
1	321	13.09	105	12.88
2	2057	83.89	698	85.64
3	74	3.02	12	1.47
Lactation start reason				
Normal calving	2426	98.94	808	99.14
Abort	26	1.06	7	0.86
Calf survival				
Survived	2155	87.89	727	89.2
Did not survived	297	12.11	88	10.8
Twinning				
Yes	14	0.57	3	0.37
No	2438	99.43	812	99.63
Calving season[c]				
Spring	543	22.15	181	22.21
Summer	664	27.08	218	26.75
Fall	571	23.29	206	25.28
Winter	674	27.49	210	25.77

[a] Categorical variable indicating which milking was used for sampling purpose by Valacta techinicians. Later, it was also used by Valacta to adjust the milk yield as well as its constituents from the first test day (ICAR, 2017).
[b] Daily milking frequency.
[c] Spring = March 20th to June 20th; Summer = June 21st to September 21st; Fall = September 22nd to December 20th; Winter = December 21st to March 19th.

ratio were evaluated in order to reduce the occurrence of overfitting and to ensure the generalization capacity of the ANN.

Only symmetric ANNs were evaluated since this architecture have shown better results compared to asymmetric architectures (Larochelle et al., 2009). In addition, the ADADELTA algorithm was used to train the model; therefore, there was no need to evaluate different learning rates (Zeiler, 2012).

The linear activation function was used in all ANN models between the last hidden layer and the output layer. In addition, all ANN evaluated had 26 neurons in the input layer, representing the variables selected by regularized regression via elastic net, and 1 neuron in the output layer, representing FT_MILK (Tables 1 and 2).

The models were constructed using the training data set defining a maximum epoch of 500 and batch size of 20 observations. During training, early stopping criterion was adopted to reduce the chance of overfitting by the ANNs and to reduce training time (Coulibaly et al., 2000). Training was allowed for n epochs until the improvement in prediction accuracy of the model, which was measured in the validation data set at the end of each epoch, was lower than 1×10^{-4} for 20 consecutive epochs. The mean absolute error (Table 3) was the statistic used to measure the accuracy of the models.

The weights of the connections between the neurons were initialized according to Glorot and Bengio (2010). Batch normalization was applied in all ANN models of the grid search along with the removal of the bias term (Ioffe and Szegedy, 2015). The training and selection of the best architecture were performed using the Keras interface (Allaire and Chollet, 2018) as implemented in R (R Core Team, 2018).

The best ANN architecture found through grid search had three hidden layers with 100 neurons in each (Fig. 1), regularization factor L1 = 1×10^{-3}, and exponential linear unit as the activation function. This model was used for further analyzes and it is from now on referred as ANN.

Table 3
Criteria used to evaluate model performance and their respective formula.

Criterion	Formula[a]
Coefficient of determination	$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$
Root mean squared error	$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n}}$
Mean absolute error	$MAE = \frac{\sum_{i=1}^{n}\mid Y_i - \hat{Y}_i\mid}{n}$
Mean percentage error	$MPE = \frac{\sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n}}}{\bar{Y}} \times 100$
Symmetric mean absolute percentage error	$SMAPE = \frac{100\%}{n}\sum_{i=1}^{n}\frac{\lvert Y_i - \hat{Y}_i\rvert}{(Y_i + \hat{Y}_i)/2}$
Standard deviation ratio	$SDr = \frac{SD_{pred}}{SD_{obs}}$

[a] Y_i = Observed milk yield on the first test day in the i^{th} animal; $\hat{Y}_i$ = Predicted milk yield on the first test day in the i^{th} animal; $\bar{Y}$ = Observed average milk yield in the first test day; n = Number of observations; DP_{pred} = Standard deviation of predicted milk yield of the first test day; DP_{obs} = Standard deviation of observed milk yield of the first test day.

2.4. Model evaluation

Six criteria were used to evaluate the performance of the models (Table 3). The efficacy of adjustment was evaluated through the coefficient of determination (R^2), while the deviation between the observed and predicted values were evaluated by root mean squared error (RMSE), mean absolute error (MAE), mean percentage error (MPE), symmetric mean absolute percentage error (SMAPE), and the ratio between predicted and observed standard deviations (SDr). The best model would have the highest R^2 and lowest RMSE, MAE, MPE, SMAPE, and SDr. In addition, 10-fold cross validation was also applied to the complete data set (training and validation data sets combined) in order to evaluate all three models.

The predictive capability of the models was evaluated as follows: since the Shapiro-Wilk test indicated that predicted and observed FT_MILK did not follow a normal distribution ($P < 0.05$), the Wilcoxon test adjusted for paired measurements was used to compare predicted and observed values. In addition, the Pearson linear correlation coefficient and the Lin's coefficient of agreement were also calculated.

The Local Interpretive Model-Agnostic Explanations (LIME) methodology proposed by Ribeiro et al. (2016) and implemented in R through the *lime* package (Pedersen and Benesty, 2018) was used to explain how the different models (MLR, RF, and ANN) integrated the independent variables to predict FT_MILK. The *lime* function was used to create explanations based on the training data set. Later, they were used in the *explain* function to create the explanations of predictions in the validation data set.

3. Results

The VIF values between the 15 variables selected by the regularized regression via elastic net ranged from 1.01 to 2.87. Therefore, it indicated that multicollinearity was not observed in our study, since all VIF values were lower than 5.00 (James et al., 2013). Correlation analysis showed that two variables had a bivariate correlation between 0.35 and 0.60, and three variables had a bivariate correlation higher than 0.60. In addition, 0.11 was the highest auto-correlation value in our study. It was observed for the time lag 1, and it decreased in the other time lags. Therefore, it did not indicate a strong auto-correlation between the FT_MILK observations. These results suggest that there was not a great overlap between the 15 variables selected by the regularized regression via elastic net.

The residual error analysis of the MLR model showed that the assumptions inherent to this methodology were not met. The homoscedasticity and the normality of the residual error (Fig. 2), as well as the independence, were all violated ($P < 0.05$).

The ANN had the best adjustment results on validation data set and 10-fold cross validation, which were higher than MLR and RF (Table 4). In addition, similar results were found between the training and validation data sets for MLR and ANN, as well as for the cross validation (Table 4). Therefore, it indicated that there was no overfitting in these two models. The opposite was observed for the RF (Table 4), which indicated the occurrence of overfitting; thus, reducing the generalization of this model.

The Pearson correlation and Lin's concordance correlation coefficients between observed and predicted values on training and validation data sets (Table 5) demonstrated a similar behavior as observed in the adjustment statistics (Table 4). The highest values were observed for RF, followed by ANN and MLR on the training data set. On the other hand, the highest values were observed for ANN followed by RF and MLR in the validation data set.

The median values of FT_MILK predicted by MLR, RF, and ANN on the validation data set were not statistically different ($P > 0.05$) from the observed, which suggests that all three models were capable to carry out FT_MILK predictions (Table 6).

The LIME methodology to interpret models assumes that the relationship between dependent and independent variables is locally linear for each prediction, and it can be modeled by a sparse linear model (Ribeiro et al., 2016). Therefore, it is possible to get the relative

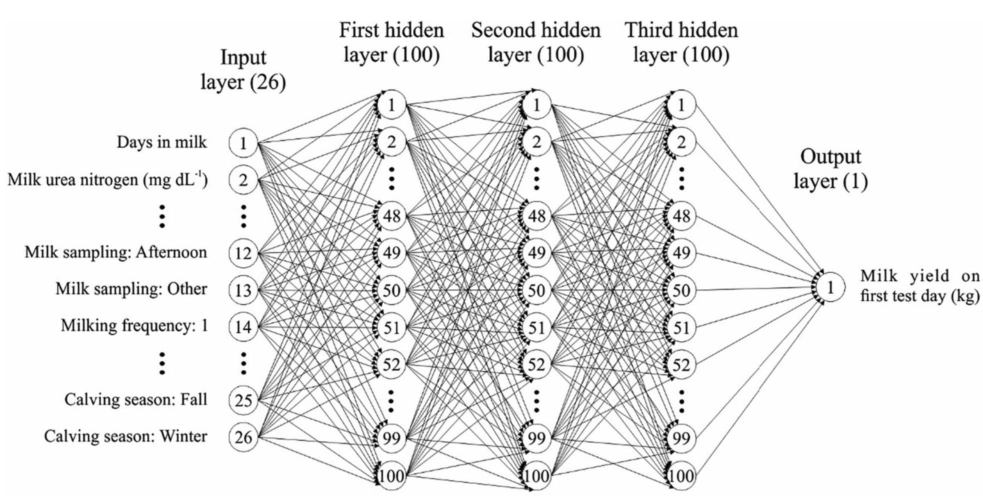

Fig. 1. The artificial neural network used in the present study containing 26 neurons in the input layer, representing the variables selected by regularized regression via elastic net (Tables 1 and 2), 100 neurons in each one of the three hidden layers, and one neuron in the output layer.

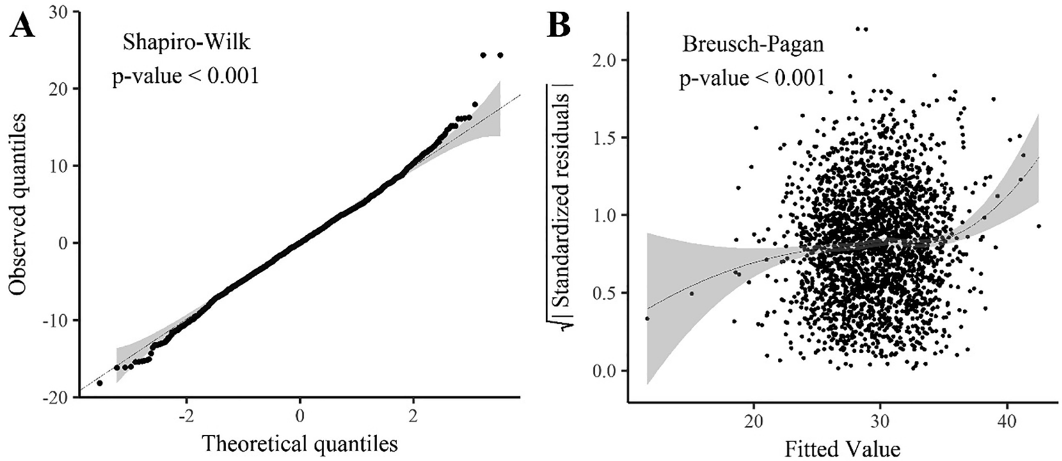

Fig. 2. A – QQ-Normal plot of the residual error of the multivariate linear regression followed by 95% confidence interval of probability (gray) and Shapiro-Wilk normality test result. B – Standardized residual error versus predicted values followed by 95% confidence interval of probability (gray) and Breusch-Pagan test result.

importance of each variable to the prediction made in addition to the coefficient of determination (R^2), which in turn indicates the quality of the interpretation (Ribeiro et al., 2016). The R^2 (mean ± standard deviation) obtained for the MLR, RF, ANN models were 0.92 ± 0.015, 0.78 ± 0.006, and 0.89 ± 0.004, respectively, indicating good quality of the interpretations.

Variables that most influenced the predictions were similar between MLR and RF, whereas ANN differed from the other two models (Fig. 3). The variable that positively influenced the most the predictions made by MLR was normal calving as the lactation start reason. On the other hand, milk β-hydroxybutyrate was the most negative influential variable on MLR predictions [Fig. 3 – A (I) and B (I)]. On RF model, milk β-hydroxybutyrate was also the one that negatively influenced the most the predictions, while average daily weight gain between 12 and 15 months of age was the one with the greatest positive influence towards FT_MILK, followed by normal calving as lactation start reason [Fig. 3 – A (II) and B (II)]. On the other hand, days in milk at test day and weight at calving were the variables with the highest positive influence on ANN model, whereas twinning was the one that most negatively influenced FT_MILK prediction of the ANN model [Fig. 3 – A (III) and B (III)].

4. Discussion

Multiple linear regression (MLR), random forest (RF), and artificial neural network (ANN) were used in the present study to predict first test day milk yield (FT_MILK) of dairy heifers. It was done so it could potentially be used as a tool to evaluate the transition period of animals in this category in a similar manner as it has been done for multiparous cows (Nordlund, 2006).

The success or failure of the transition programme have been successfully represented by FT_MILK. Heuer et al. (1999) reported that milk yield and fat to protein ratio better predicted the occurrence of transition problems than change in body weight before and after calving. In addition, metabolic disorders, which have a higher incidence around parturition (Østergaard and Gröhn, 1999), negatively affects FT_MILK (Gantner et al., 2016) and the negative effect is carried out throughout the whole lactation (Drackley, 1999). Thus, milk production during the transition period reflects the transition program employed and it justifies the use of FT_MILK as a means to evaluate it.

All three models evaluated in the presented study showed a FT_MILK predictive capacity. To the best of our knowledge, this is the first study that aimed at creating a model to predict FT_MILK of dairy heifers. Therefore, it was not possible to compare the predictive performance of the models created in our study with those of other studies, even though comparisons between models is difficult due to differences in data sets and variables used to create the model (Murphy et al., 2014).

The results reported here indicated the occurrence of overfitting of the RF model on the training data set, which in turn reduced its generalization capacity as observed in the results obtained in the validation data set and 10-fold cross validation. Breiman (2001b) suggested that RF does not suffer from overfitting while evaluating its performance in benchmarking data sets from the University of California, Irvine (UCI) repository. However, Segal (2004), while evaluating the same data sets,

Table 4
Results of multivariate linear regression, random forest, and artificial neural network models on training (n 2452) and validation (n = 815) data sets as well as 10-fold cross validation and 95% confidence interval.

Item [a]	Multivariate linear regression			Random forest			Artificial neural network		
	Training	Validation	Cross validation (95% CI) [b]	Training	Validation	Cross validation (95% CI) [b]	Training	Validation	Cross validation (CI 95%) [b]
R^2	0.30	0.30	0.29 (0.27–0.32)	0.93	0.32	0.30 (0.28–0.33)	0.34	0.32	0.32 (0.28–0.36)
RMSE	5.04	5.10	5.08 (4.89–5.28)	0.25	5.07	5.05 (4.90–5.21)	4.91	5.02	4.98 (4.77–5.19)
MAE	3.94	3.99	3.98 (3.88–4.09)	1.73	3.94	3.94 (3.87–4.02)	3.80	3.89	3.88 (3.75–4.00)
MPE	17.07	17.33	17.23 (16.60–17.87)	7.64	17.21	17.14 (16.63–17.65)	16.64	17.05	16.89 (16.32–17.46)
SMAPE	13.64	13.99	13.84 (13.46–14.22)	6.17	13.77	13.71 (13.41–14.01)	13.16	13.62	13.48 (13.15–13.81)
SDr	0.55	0.52	0.55 (0.53–0.57)	0.70	0.48	0.50 (0.49–0.52)	0.54	0.54	0.54 (0.51–0.58)

[a] R^2 = Coefficient of determination; RMSE = Root mean squared error; MAE = Mean absolute error; MPE = Mean percentage error; SMAPE = Symmetric mean absolute percentage error; SDr = Standard deviation ratio.

Table 5
Pearson correlation coefficient (r) and Lin's concordance correlation coefficient (ρ) followed by 95% confidence interval of probability (95% CI) obtained between the predicted and observed values for the training (n = 2452) and validation (n = 815) data sets.

Model	Training			Validation		
	r	p – value	ρ (CI 95%)	r	p – value	ρ (CI 95%)
Multivariate linear regression	0.55	< 0.001	0.46 (0.44–0.49)	0.55	< 0.001	0.45 (0.40–0.49)
Random forest	0.96	< 0.001	0.91 (0.90–0.91)	0.56	< 0.001	0.44 (0.40–0.48)
Artificial neural network	0.58	< 0.001	0.49 (0.46–0.51)	0.57	< 0.001	0.48 (0.43–0.52)

Table 6
Comparison between predicted and observed median first test day milk yield in the validation data set (n = 815).

Model	Milk yield		p – value
	Observed (Median ± absolute deviation)	Predicted (Median ± absolute deviation)	
Multivariate linear regression	29.44 ± 5.93	29.53 ± 3.40	0.61
Random forest	29.44 ± 5.93	30.09 ± 3.03	0.24
Artificial neural network	29.44 ± 5.93	29.58 ± 3.39	0.37

concluded that tree-based models would hardly suffer from overfitting on that data sets and it might not be true for different data. Our results suggest that RF models suffers from overfitting, even though the real reason behind its occurrence is not yet known (Strobl et al., 2009).

The best model in our study was the ANN. Additionally, ANN is more flexible than MLR since it does not make assumptions regarding data distribution, such as homoscedasticity and normality of the residual error (Adamczyk et al., 2016), which were both not met in the present study.

Milk β-hydroxybutyrate was the variable that negatively affected the most the prediction of FT_MILK done by MLR and RF models. Milk β-hydroxybutyrate can be used as an indicator of ketosis in dairy cows (Denis-Robichaud et al., 2014), which is one of most prevalent metabolic disorders during the transition period with an incidence rate ranging from 26 to 56% (McArt et al., 2012). Specially on primiparous, Santschi et al. (2016) reported an average prevalence of 18.8% and the highest prevalence (30.3%) was observed in the first week after calving. Ketosis is considered to be a risk factor for other metabolic disorders and infectious diseases (Gröhn et al., 1989; Suthar et al., 2013) as well as poor reproductive performance even at subclinical levels (Rutherford et al., 2016). In addition, dairy heifers that suffer from ketosis produce, on average, 8 to 10% less milk compared with healthy animals (Santschi et al., 2016). Thus, the negative effect attributed to milk β-hydroxybutyrate by both the MLR and RF models while predicting FT_MILK is biologically justifiable.

Body weight at calving and days in milk at the test day were the variables that most positively influenced the prediction of FT_MILK done by the ANN model, which means that the higher the body weight at calving and days in milk at first test day, the higher would be the predicted milk yield for the first test day. Based on the expected lactation curve, a daily increase in milk production is observed during the first weeks after calving until it reaches the peak yield and gradual reduction is observed towards the end of the lactation. Thus, it would be expected a positive relationship between days in milk and FT_MILK as observed in the ANN model, even though the same was not observed on MLR and RF models. In addition, weight at calving has been previously shown to be positively related to productive performance of dairy heifers (Macdonald et al., 2005), biologically justifying the positive relationship to FT_MILK attributed by the ANN model.

On the other hand, twinning was the variable that most negatively affected the prediction of FT_MILK done by the ANN model. Although the prevalence of twinning is low in dairy cows, it has a great negative economic effect on the dairy activity (Mur-Novales et al., 2018). It has also been associated with an increased risk of developing reproductive problems such as metritis and retained placenta as well as ketosis (Nielen et al., 1989) and dystocia (Hossein-Zadeh, 2010). Consequently, FT_MILK tends to be smaller (Hossein-Zadeh, 2013).

The present work has demonstrated that it is possible to predict first test day milk yield of dairy heifers based on DHI metrics. However, only data from Holstein heifers were used to train and evaluate our models, making it necessary to create and evaluate models for breeds other than Holstein. Future work could also focus on evaluating the variation between herds in order to identify trends that could be included in the models. Herd factor has been found significant in evaluating the immune response of transition cows (Zecconi et al., 2018), risk of abortion (Keshavarzi et al., 2017), metabolic disorders, reproductive performance, and culling rate (Heuer et al., 1999). Thus, it is reasonable to hypothesize that its inclusion during the modeling could further improve the predictive capacity of the ANN, since it was the best model in the present study. In addition, the FT_MILK predictions made by the ANN still have to be validated as a tool to evaluate the transition period of dairy heifers. Such approach has been previously validated for multiparous cows (Nordlund, 2006) and our work was the first step towards developing a similar tool for primiparous cows.

5. Conclusion

Fifteen DHI variables were identified as the most influential on first test day milk yield (FT_MILK) of Holstein dairy heifers. Multivariate linear regression (MLR), random forest (RF) and artificial neural network (ANN), which were the three modelling methods evaluated in the present study, were able to predict FT_MILK. However, the ANN model was more accurate in its predictions compared to MLR and RF models, especially since overfitting was observed in the RF. Our study also suggests that ANN is a more viable alternative for modelling DHI metrics compared with conventional techniques such as MLR. In addition, the most influential variables differed between all three models evaluated.

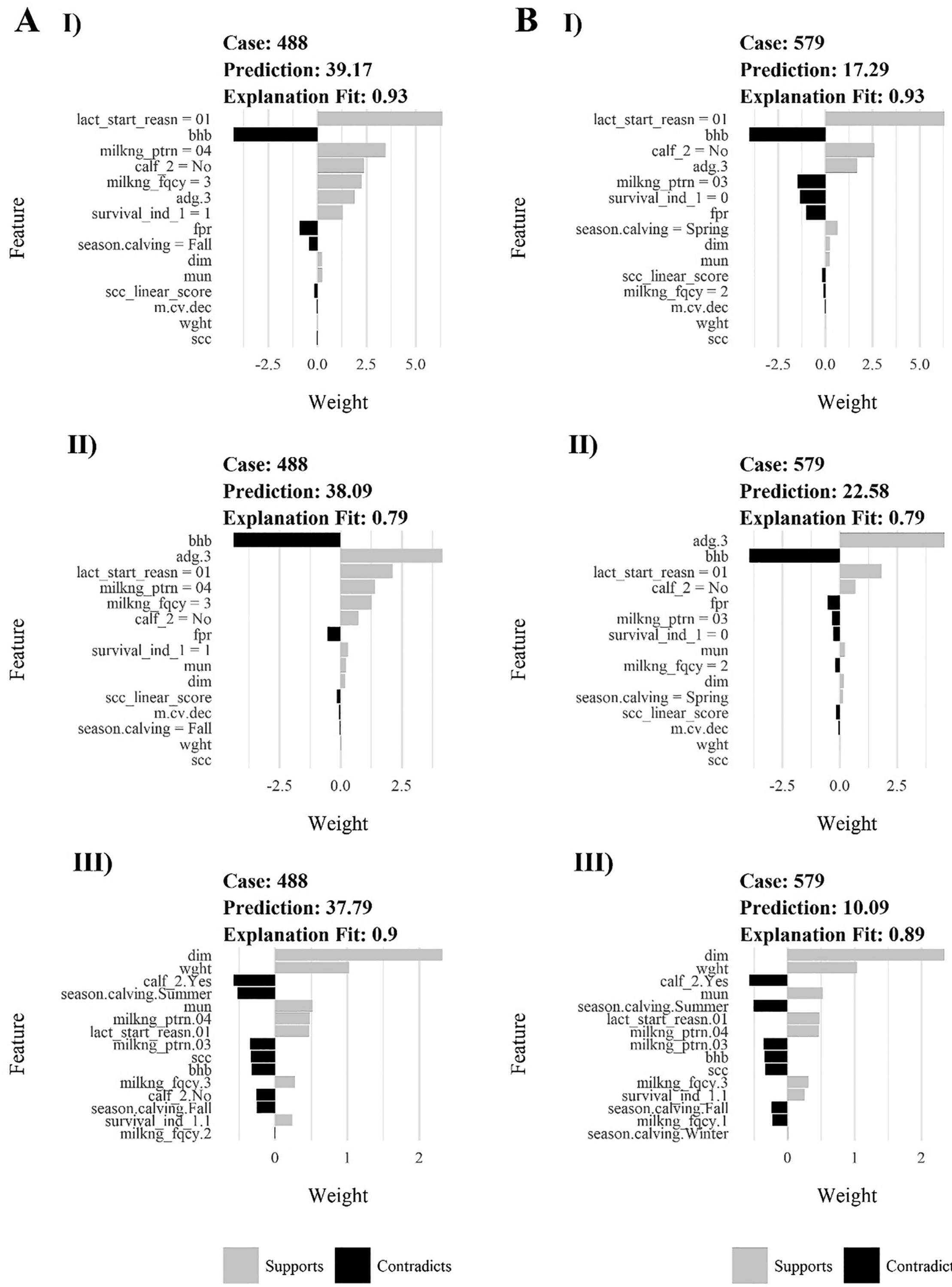

Fig. 3. The relative importance of the variables in the greatest (A) and lowest (B) predictions made by multivariate linear regression (I), random forest (II), and artificial neural network (III) models. lact_start_reasn = Lactation start reason, code 1: Normal calving; bhb = Milk β-hydroxibutirate (mmol L^{-1}); milkng_ptrn = Milk sampling, code 03: 24 h sampling, code 04: Other sampling; calf_2.No = Did not occur twinning; calf_2.Ye =: Occurred twining; milkng_fqcy = Milking frequency, code 1: Once a day, code 2: Twice day, code 3: Three times a day; adg.3 = Average daily gain between 12 and 15 months old (Kg $animal^{-1}$ day^{-1}); survival_ind_1 = First calf survival, code 0: Died, code 1: Survived; fpr = Fat to protein ratio; season.calving = Lactation start season (spring, summer, fall, and winter); dim = Days in milk at test day; mun = Milk urea nitrogen(mg dL^{-1}); scc = Somatic cell count (10^3 cells mL^{-1}); scc_linear_score = Somatic cell count linear score; m.cv.dec = Month of calving; wght = Body weight at calving (Kg).

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Declaration of Competing Interest

The authors have no conflict of interest to declare.

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